FLOORPLANNING FOR EDA • LAB 07

Simulated Annealing Floorplan Search

Toy cost adds one thousand when a packing is illegal

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learn_floorplanning — 07 simulated annealing fp

ALGORITHM SPECIFICATION

Pseudocode Workflow

```
INPUT: Modules, Temp T_0, Cooling Rate alpha
cost = deadspace + HPWL + 1000 * (!legal)

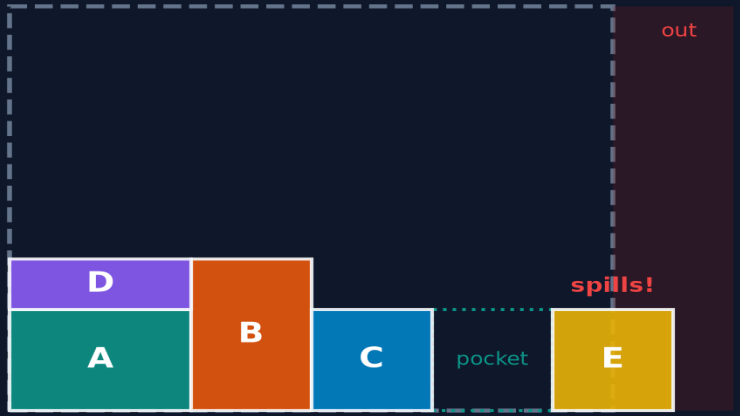
WHILE T > T_min DO:
  S_cand = saSwap(S_curr)  // Swap origin coordinates
  delta_C = cost(S_cand) - cost(S_curr)
  IF delta_C <= 0 OR rand() < e^(-delta_C/T):
    S_curr = S_cand
    IF legal(S_curr) AND cost(S_curr) < cost(S_best):
      S_best = S_curr
  T = T * alpha
```

STATE TRANSITION

Bad Seed vs. Golden Packing

Step 1: Bad Overflow Seed

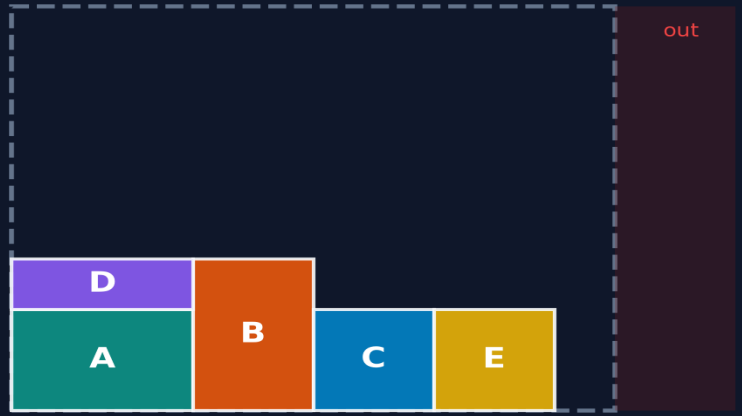
Illegal placement incurs penalty tax



Cost: ~1044 | Legality: False

Step 2: Golden Legal Packing

Zero overlap; cost drops below penalty floor



Cost: ~36 | Legality: True

DEBUGGING & CHECKLIST

Implementation Pitfalls

- **Accepting illegal states without penalty:** Leaves solver stuck in invalid layouts.
- **Forgetting to rescore:** Evaluating Metropolis state with stale costs.
- **Quenching (Cooling too fast):** Traps search in local minima prematurely.
- **Missing the empty-pocket move:** Pairwise swaps cannot fill the hole.